Internet Protocol

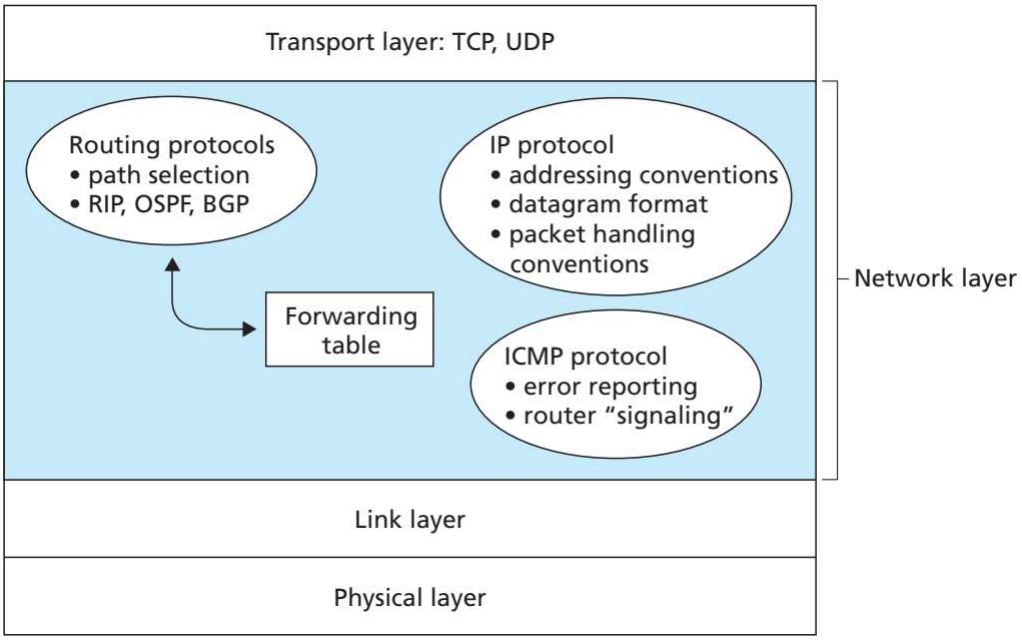
Internet Protocol Version 4 (IPv4)

Internet Protocol Version 6

Acknowledgement: Some of the slide contents are taken from the slides shared by the author of the book **Computer Networking: A Top-Down Approach 9th Edition, Jim Kurose, Keith Ross**. Some of the figures are also taken from the same book. Authors are sincerely acknowledged

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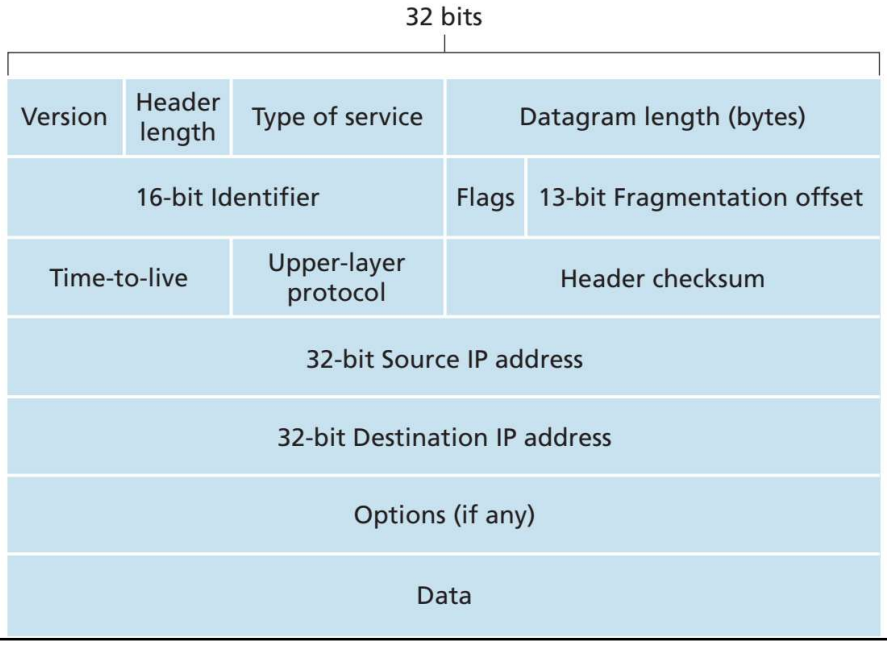
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Datagram Format

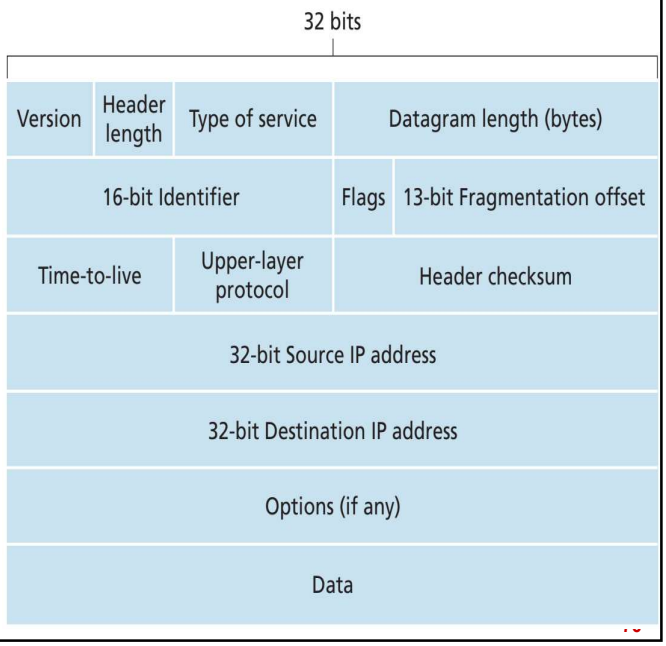


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Datagram Format

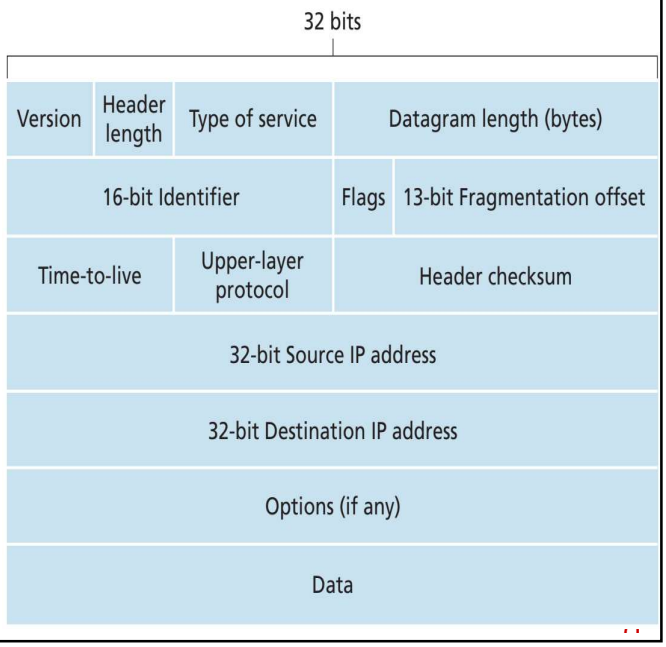
- **Datagram is network layer packet**
- **Version (4 bits)**
  - **IP Protocol version of the datagram-IPv4, IPv6**
    - **Rest of the datagram is interpreted accordingly**
- **Header Length(4 bits)**
  - **Where the actual data of the datagram begins**
  - **Typical length – 20 bytes**



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Datagram Format

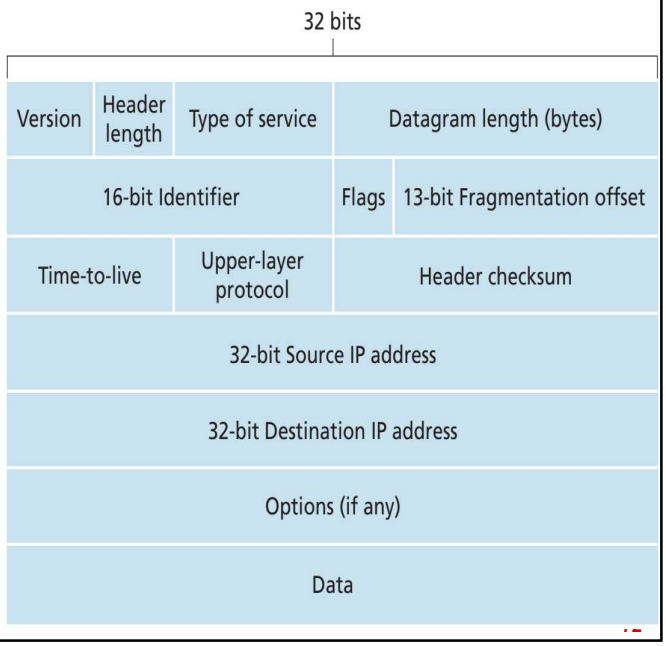
- **Type of service (8 bits)**
  - **Denote type of datagram**
    - **Low delay**
    - **High throughput**
    - **Reliability**
  - **Upper 6 bits– Assign different levels of priority to packets**
  - **Lower 2 bits-Explicit congestion notification**
    - **Allows routers to notify end-hosts of network congestion without dropping packets**



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Datagram Format

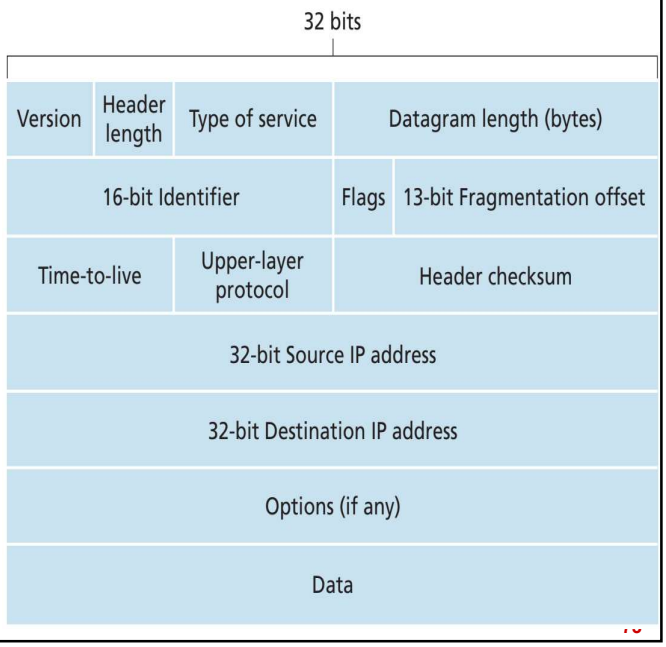
- **Datagram length (16 bits)**
  - **Total length of IP datagram**
    - Measures in bytes
    - Maximum size 65535 bytes
    - Generally, not longer than 1500 bytes
- **Identifier, flags, fragmentation offset**
  - **Concerned with IP fragmentation**
    - Typically, not supported in IPv6



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IP Fragmentation

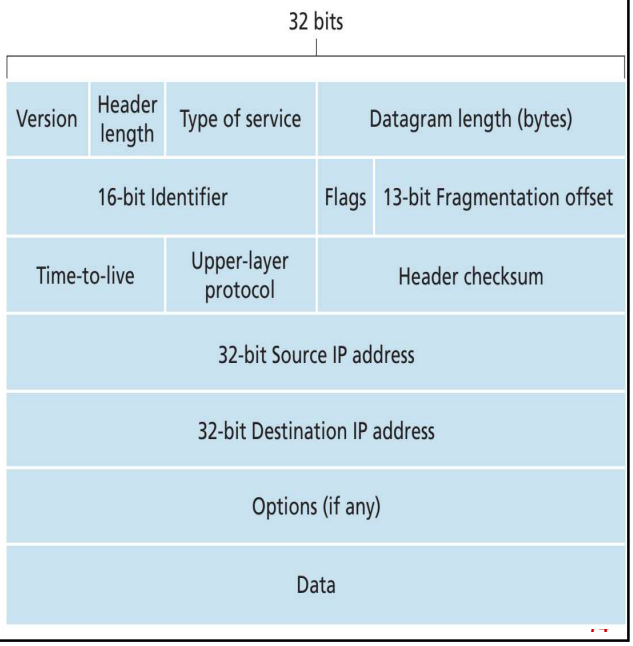
- **Identifier, flags, fragmentation offset**
  - **Link-layer frame can carry a specific size frame**
    - Ethernet -1500 bytes
    - Wide-area links—576 bytes
    - **Maximum transmission unit (MTU)**
      - **Maximum amount of data a link-layer frame can carry**
  - **MTU – hard limitation on the size of IP datagram to be carried from one router to another**
    - **May be every IP datagram may be sized accordingly**
      - **Links between the routers along the paths have different MTUs**



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IP Fragmentation

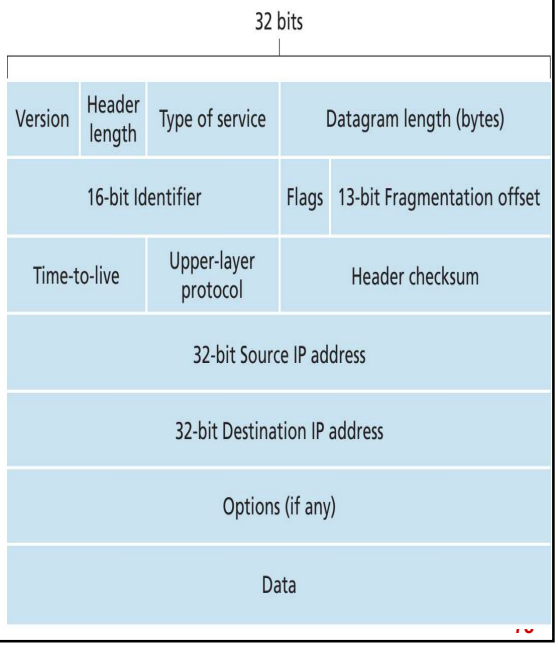
- **Identifier, flags, fragmentation offset**
  - **Link-layer frame can carry a specific size frame**
  - **MTU – hard limitation on the size of IP datagram to be carried from one router to another**
    - **May be every IP datagram may be sized accordingly**
      - **Links between the routers along the paths have different MTUs**
  - **Fragment the data in IP datagram into smaller datagrams**
    - **Smaller datagram is called a fragment**
  - **Fragments need to be reassembled when they reach the transport layer at the destination host**
    - **Router?**
    - **End system?**



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IP Fragmentation

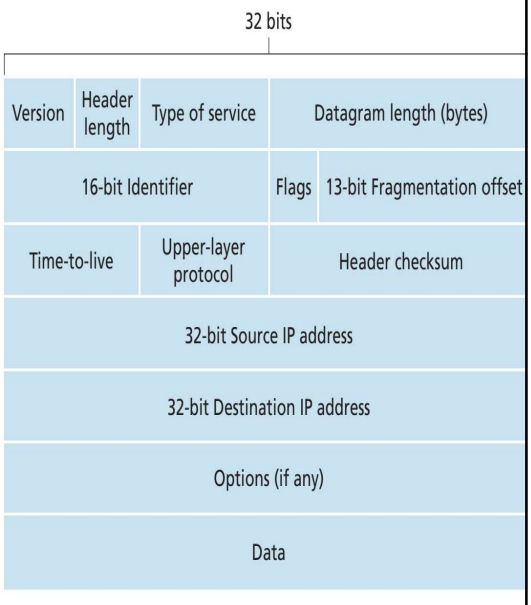
- **Identifier, flags, fragmentation offset**
  - **Fragment the data in IP datagram into smaller datagrams**
    - **Smaller datagram is called a fragment**
  - **Fragments need to be reassembled when they reach the transport layer at the destination host**
    - **Router-burden in addition to everything else done**
    - **End system—IPv4**
  - **Destination host receiving the series of datagrams need to reassemble them into a large datagram**
    - **Series received are part of a large datagram (fragments)**
    - **The last fragment received and previous 'n' fragments must be put together**
    - **To facilitate this these fields are used**



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IP Fragmentation

- **Identifier, flags, fragmentation offset**
  - **Every datagram sent out by sending host attaches an identification number, source and destination addresses**
    - Identification number is incremented for every new datagram sent out
  - When a router fragments a datagram due to MTU limitation, it will use the same identification number in the newly created datagram along with the source and destination IP address
  - **Destination can examine the identification number of series of datagram received to notice the set of fragments of original datagram**
    - **What if some fragments do not arrive at the destination at all**
    - **To ensure that the last fragment received – flag field is checked**
      - **Flag=0 – last segment, Flag=1 – not last segment**

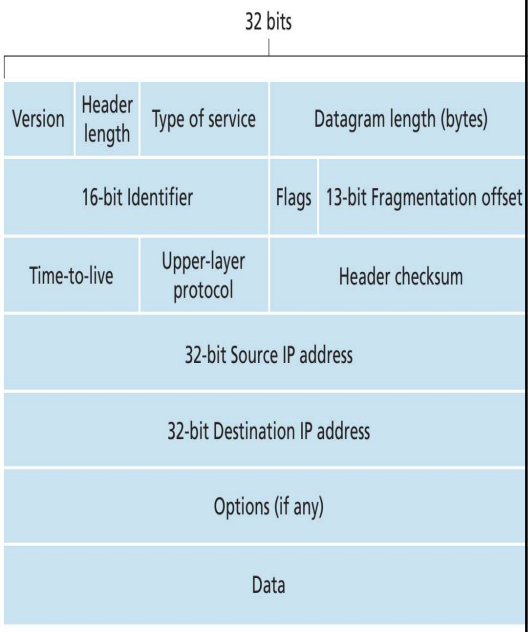


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IP Fragmentation

- **Identifier, flags, fragmentation offset**
  - **Destination can examine the identification number of series of datagram received to notice the set of fragments of original datagram**
    - **What if some fragments do not arrive at the destination at all**
      - **To ensure that the last fragment received – flag field is checked**
        - » **Flag=0 – last segment, Flag=1 – not last segment**
      - **If some fragments in between are missing**
        - » **Offset field to specify where in the original datagram current fragment fits**
    - **Example**
      - **A datagram of 4000 bytes (20 bytes IP header and 3980 bytes of IP payload(data)) arrives at a router and must be forwarded to a link with an MTU of 1500 bytes**

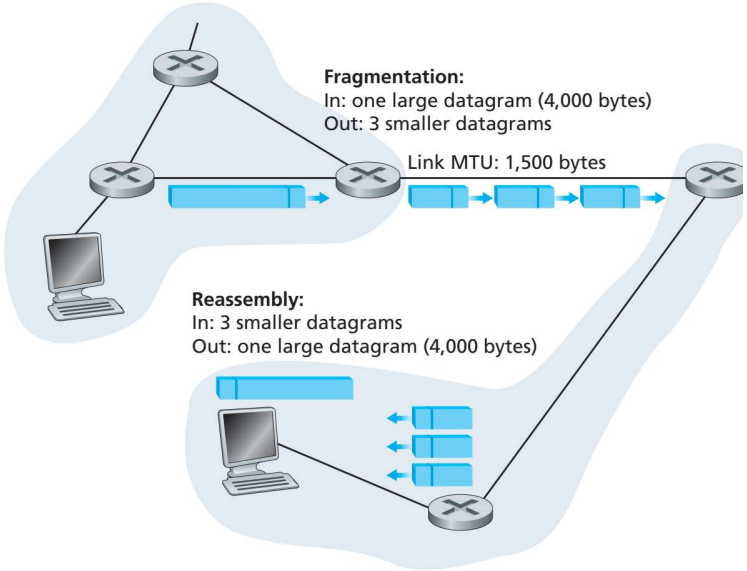


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IP Fragmentation

- **Example**
  - **A datagram of 4000 byte (20 bytes IP header an 3980 bytes of I payload(data)) arrives at router and must b forwarded to a link with a MTU of 1500 bytes**



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IP Fragmentation

- **Example**
  - **A datagram of 4000 bytes (20 bytes IP header and 3980 bytes of IP payload(data)) arrives at a router and must be forwarded to a link with an MTU of 1500 bytes**

Fragment	Bytes	ID	Offset	Flag
1st fragment	1,480 bytes in the data field of the IP datagram	identification = 777	offset = 0 (meaning the data should be inserted beginning at byte 0)	flag = 1 (meaning there is more)
2nd fragment	1,480 bytes of data	identification = 777	offset = 185 (meaning the data should be inserted beginning at byte 1,480. Note that 185 · 8 = 1,480)	flag = 1 (meaning there is more)
3rd fragment	1,020 bytes (= 3,980–1,480–1,480) of data	identification = 777	offset = 370 (meaning the data should be inserted beginning at byte 2,960. Note that 370 · 8 = 2,960)	flag = 0 (meaning this is the last fragment)

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IP Fragmentation-- Costs

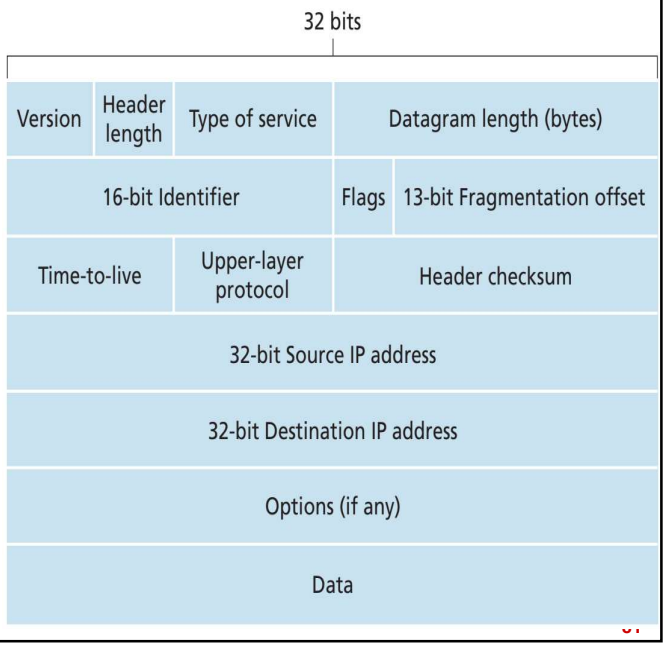
- **Complicates routers and end systems**
  - **Need to reassemble the fragments**
- **Fragmentation can be used to create DoS kind of attacks**
  - **Attacker sends series of unexpected fragments**
    - **A stream of fragments sent to target host**
      - **None of them having offset as zero**
      - **Overlapping offset**

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Datagram Format

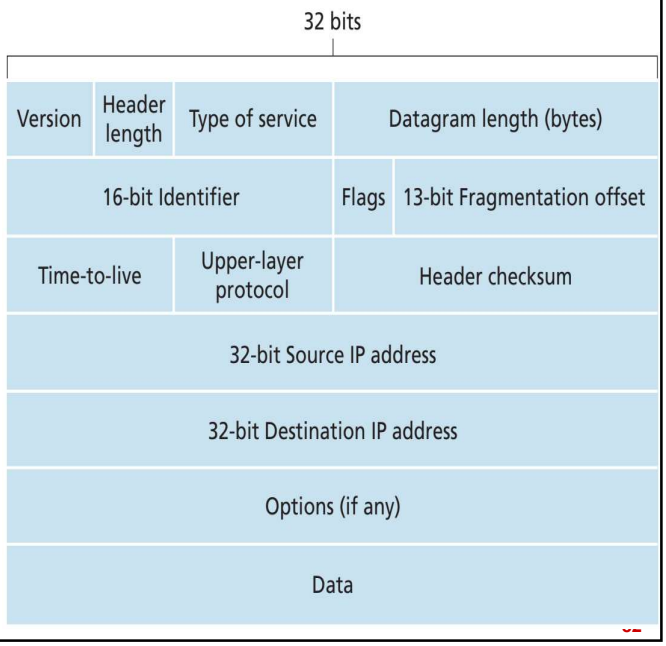
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  - **Total length of IP datagram**
    - Measures in bytes
    - Maximum size 65535 bytes
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    - Typically, not supported in IPv6



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Datagram Format

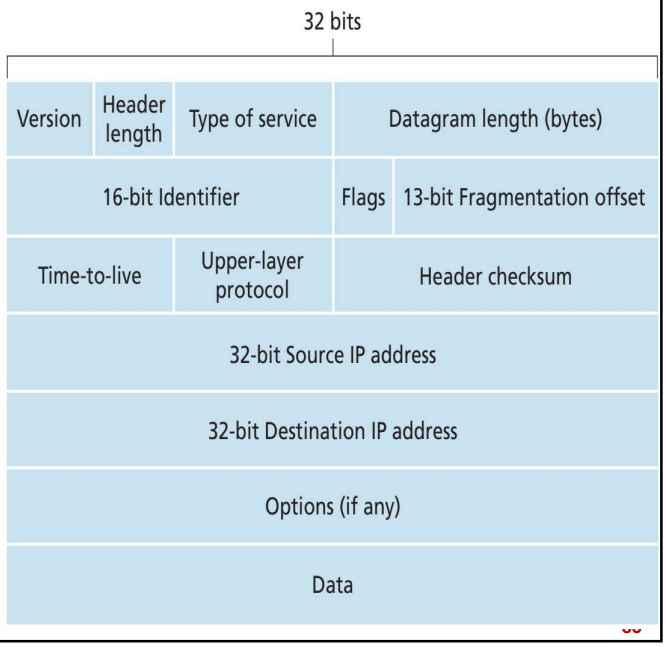
- **Time-to-live (TTL) (8 bits)**
  - **Remaining maximum hops for the packet**
    - Datagram do not circulate forever in network
    - Decremented every time a datagram passes through a router
    - TTL=0, datagram must be dropped
- **Upper layer protocol**
  - 8 bits
  - **Specific transport layer protocol to which data portion of the datagram to be passed**
    - TCP, UDP



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Datagram Format

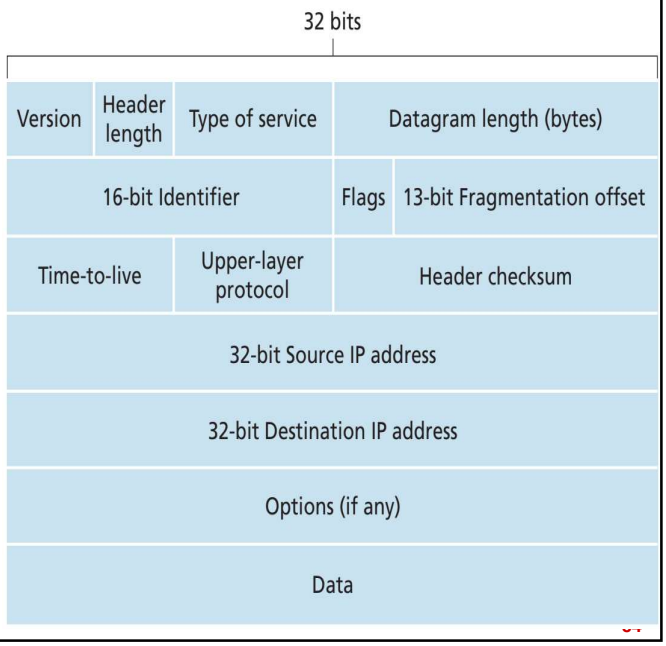
- **Header checksum (16 bits)**
  - **Aids routers in detecting bit errors in received datagrams**
  - **Header checksum**
    - Every 2 bytes in the header is treated as a number and summed using 1s complement arithmetic
      - Stored in this field in the datagram
    - Router computes the checksum as above and compares what is come in the datagram
    - Router to discard the packet if any mismatch occurs
  - **When to compute the checksum?**
    - **Every router– TTL field varies**



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Datagram Format

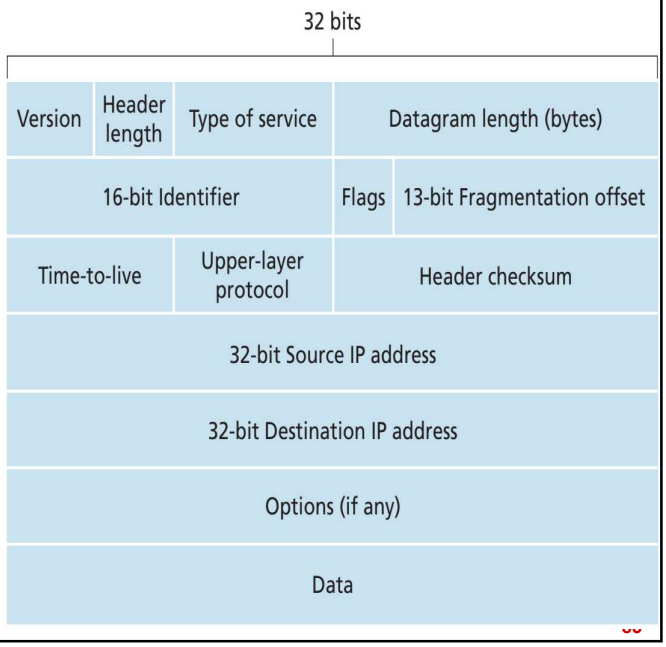
- **Source IP Address (32 bits)**
- **Destination IP Address (32 bits)**
- **Options**
  - **To extend IP header if needed**
    - Rarely extended
  - **Field not typically used**
    - May make it difficult to find where exactly the actual data starts
    - Processing at router becomes a problem
- **Data**
  - **Transport layer segment to be delivered to destination**
    - **TCP or UDP**



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Datagram Format

- Datagram is network layer packet
- Version (4 bits)
  - IP Protocol version of the datagram-IPv4, IPv6
    - Rest of the datagram is interpreted accordingly
- **Header Length(4 bits)**
  - **Where the actual data of the datagram begins**
  - **Typical length – 20 bytes**
    - **Assuming that no options are there**



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